Consider a state space $S = \{0, 1, 2\}$.

We can define a probability distribution on S by listing 3 probabilities p_0, p_1, and p_2 with $p_0 + p_1 + p_2 = 1$.

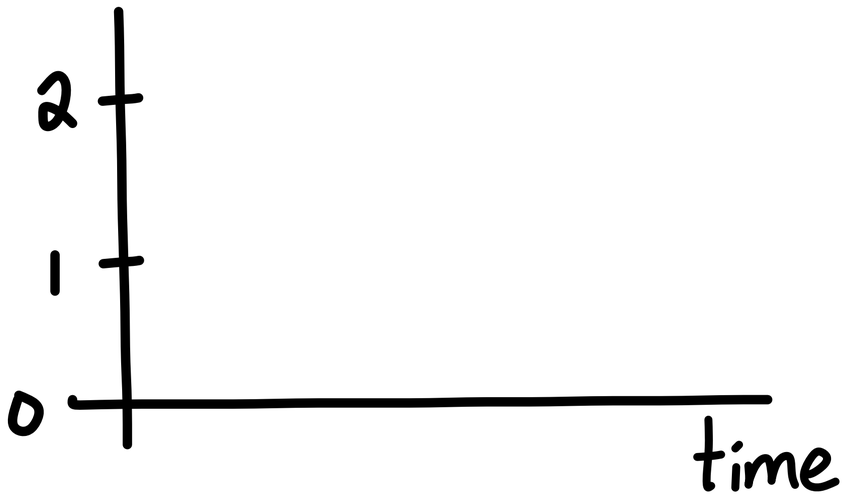
We are going to look at a "special" distribution and call these probabilities π_0, π_1, and π_2.

We will refer to the entire distribution (the entire collection of probabilities) as π.

So far, we have been starting our Markov chains at arbitrary values.

(Example: Let $X_0 = 0$.)

$S = \{0, 1, 2\}$



Consider starting

- at state 0 w.p. $\pi_0 = 1/2$
- at state 1 w.p. $\pi_1 = 1/4$
- at state 2 w.p. $\pi_2 = 1/4$

<u>Question</u>: What is the probability distribution for where the chain will be at time 1 ?

$$P(X_1 = j) = \sum_{i=0}^{2} \underbrace{P(X_1 = j \mid X_0 = i)}_{P_{ij}} \cdot \underbrace{P(X_0 = i)}_{\pi_i}$$

What if we could find a starting distribution π such that the distribution after one time-step is still π ?

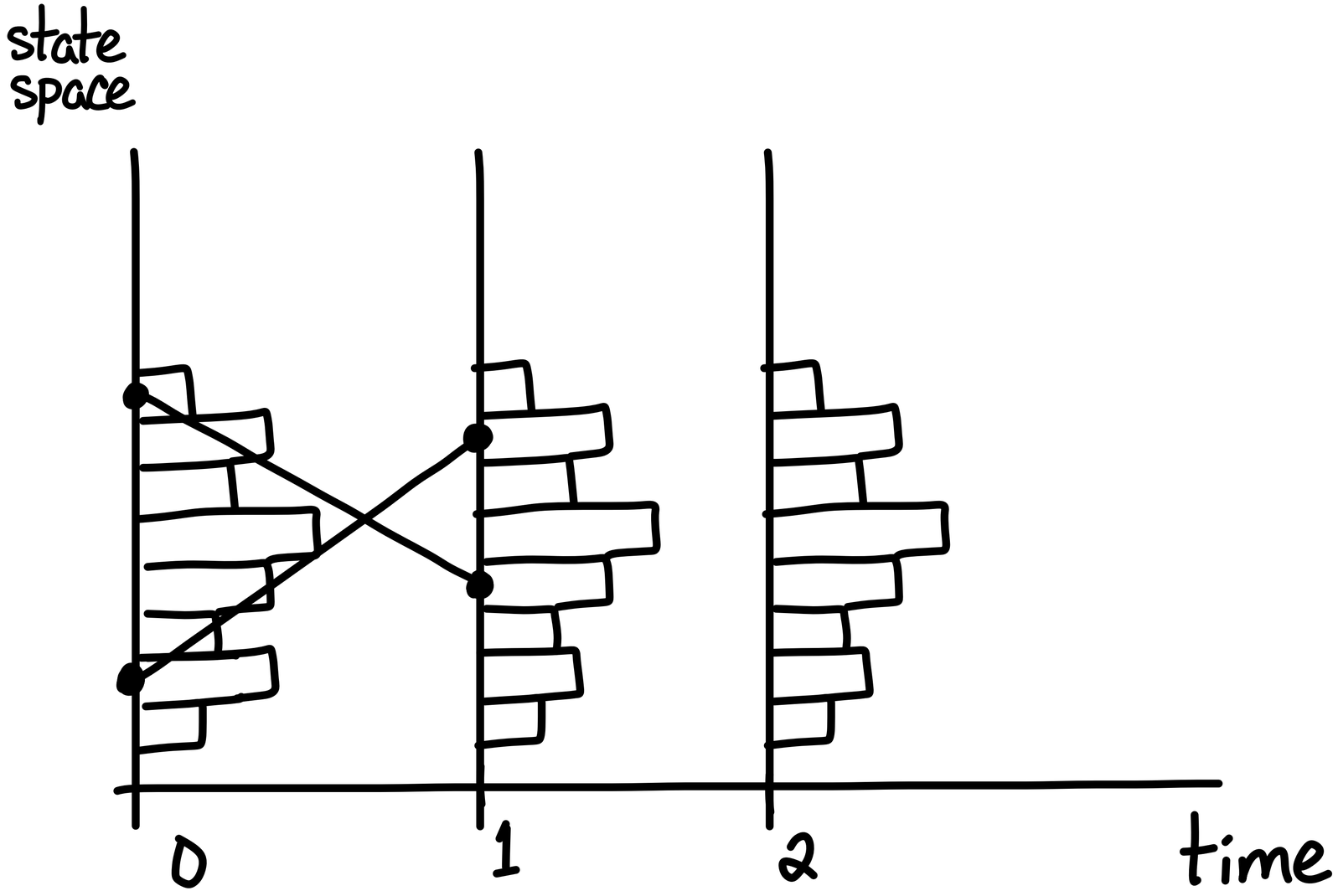
i.e. $$\pi_j = \sum_{i=0}^{2} \pi_i P_{ij}$$

Such a distribution is known as a "stationary distribution".

<u>Definition</u>: Let $\{X_n\}$ be a Markov chain on a state space S. A distribution π on S is a <u>stationary distribution</u> for the chain if

$$\pi_j = \sum_{i \in S} \pi_i P_{ij}$$

← the stationary equation


state space
0
1
2
time

⊛ When a chain is "started stationary", it will maintain this distribution at all fixed time points.

So, we actually have

$$\pi_j = \sum_{i \in S} \pi_i P_{ij}^{(n)}$$

as well!

<u>Proof</u>: (by induction)

- If π is stationary, we have

$$\pi_j = \sum_{i \in S} \pi_i P_{ij} \qquad (*)$$

- Suppose that $\pi_j = \sum_{i \in S} \pi_i P_{ij}^{(n)} \qquad (**)$

- Want to show that $\pi_j = \sum_{i \in S} \pi_i P_{ij}^{(n+1)}$

- $$\sum_{i \in S} \pi_i P_{ij}^{(n+1)} \overset{C.K.}{=\!=} \sum_{i \in S} \pi_i \sum_{k \in S} P_{ik}^{(n)} \cdot P_{kj}$$

$$= \sum_{k \in S} \sum_{i \in S} \pi_i P_{ik}^{(n)} \cdot P_{kj}$$

$$= \sum_{k \in S} P_{kj} \underbrace{\sum_{i \in S} \pi_i P_{ik}^{(n)}}_{\pi_k \text{ by } (**)}$$

$$= \sum_{k \in S} \pi_k P_{kj} = \pi_j \quad \text{by } (*) \quad //$$